

User guide

App Overview

CMSVS is a hosted document validation system available at <https://jmethew765-ml4.hf.space/>. It helps users:

- extract structured information from a single document,
- compare two documents to check whether key values match,
- build or adjust extraction rules for new document types,
- and verify extraction results against ground truth data.

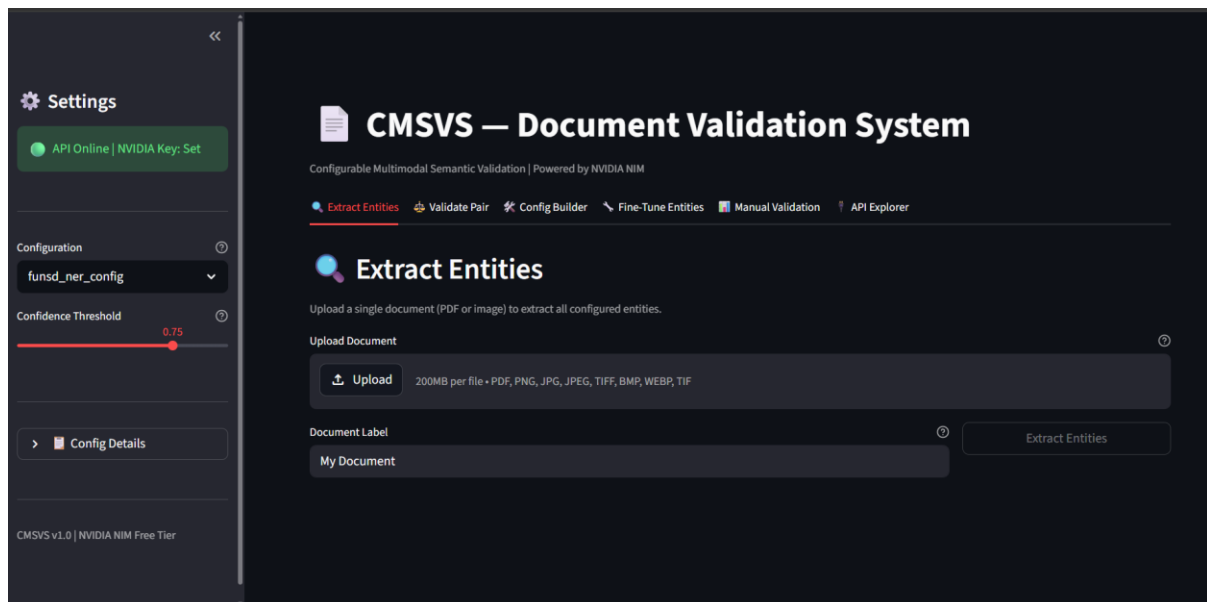
This is a user-facing app with a sidebar for settings plus tabs for extraction, validation, configuration, and evaluation.

Input Description

Users provide input by uploading files in the app interface.

Supported document inputs:

- PDF
- PNG
- JPG / JPEG



Main input methods:

- **Extract Entities:** upload one document and give it a label
- **Validate Pair:** upload two documents, one as Document A and one as Document B
- **Config Builder:** define extraction fields by form or CSV

- **Fine-Tune Entities:** select a config and section, then edit descriptions
- **Manual Validation:** upload ground truth JSON and solution output JSON for accuracy checking

The sidebar also lets users:

- confirm the API is online and the NVIDIA key is set
 - choose which config to use
 - adjust the confidence threshold for review warnings
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Output Description

The app returns clear, user-friendly results depending on the chosen task.

Extraction output:

- list of extracted fields from the uploaded document
- review flags for values that need human attention

Validation output:

- comparison of Document A and Document B
- match/mismatch summary
- full validation report or M2 ground truth format, depending on the selected option

Config Builder output:

- generated YAML config file
- extraction instructions in Markdown

Manual Validation output:

- evaluation metrics based on ground truth JSON and solution output JSON
 - accuracy feedback for extraction performance
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Step-by-Step Instructions

1. Launch the App

- Open <https://jmethew765-ml4.hf.space/>
- Confirm the left sidebar shows **API Online | NVIDIA Key: Set**

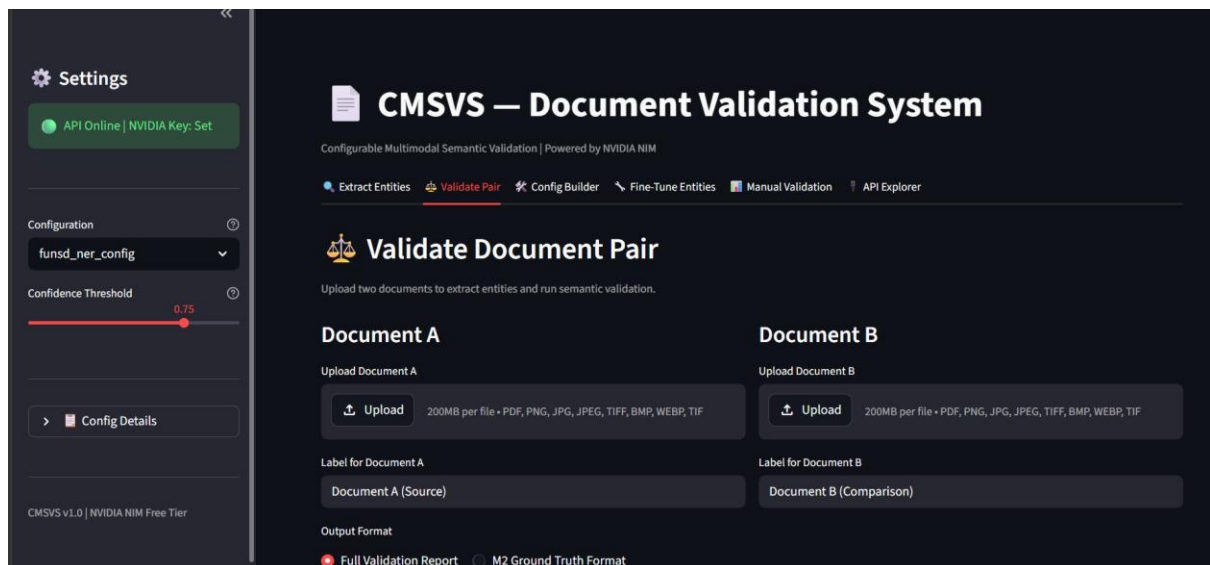
2. Configure the App

- In **Settings**, select the active configuration from Configuration
- Adjust the **Confidence Threshold** slider if needed

3. Extract Entities

- Go to the **Extract Entities** tab
- Upload a single PDF or image file
- Enter a document label if desired
- Click **Extract Entities**
- Review the extracted field list and any review warnings

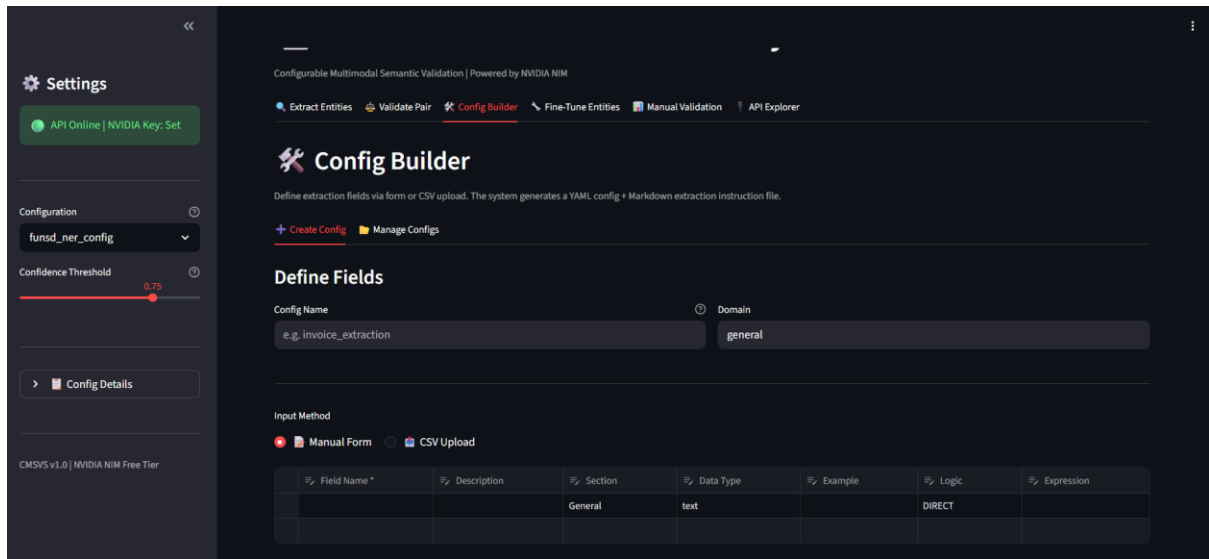
4. Validate a Document Pair



- Go to the **Validate Pair** tab
- Upload **Document A** and **Document B**
- Enter labels for both documents
- Choose either:
 - **Full Validation Report**
 - **M2 Ground Truth Format**
- Click **Run Validation**

- Review the output for matching or mismatching values

5. Build or Edit Configurations

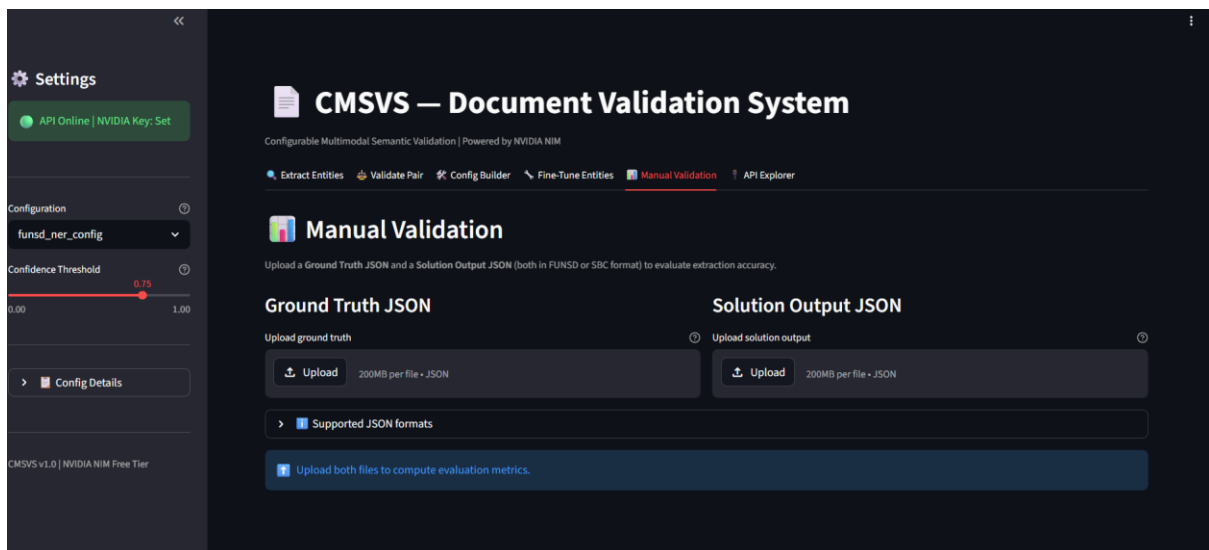


- Go to the **Config Builder** tab
- Create a new config or manage existing ones
- Define fields, descriptions, sections, and example values
- Generate a config file and extraction instructions

6. Fine-Tune Entity Descriptions

- Go to **Fine-Tune Entities**
- Select a config and section
- Edit descriptions and example values for entities
- Upload a test document to preview how the system extracts values with updated text

7. Check Accuracy with Ground Truth



- Go to **Manual Validation**
- Upload a **Ground Truth JSON**
- Upload a **Solution Output JSON**
- Use this tab to compare extraction output against known truth data

8. Explore API Endpoints

- Use the **API Explorer** tab to see the backend API capabilities
 - This is useful for users who want to understand or test the service directly
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Troubleshooting

If the app does not load:

- refresh the page
- check your internet connection
- wait a moment and try again

If the app shows **API Offline** or a missing key:

- the service may not be connected correctly
- reload the page
- the deployment should automatically reconnect when the backend is available

If uploads fail:

- confirm file type is one of the supported formats
- try a smaller file if the upload is large
- re-select the file and upload again

If validation does not run:

- make sure both Document A and Document B are uploaded
- make sure a configuration is selected
- check the confidence threshold and try again

If results look incorrect:

- use **Manual Validation** with ground truth JSON to verify
 - adjust the active config or entity descriptions in **Config Builder** or **Fine-Tune Entities**
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Simple Use Case

1. Open the app.
2. Select `funsd_ner_config` or another available config.
3. Upload a single document in **Extract Entities** to see extracted values.
4. Upload two documents in **Validate Pair** to compare them.
5. Use **Manual Validation** if you have expected results in JSON form.

This documentation is written for general users to understand what the app does, how to use it, and how to troubleshoot basic problems.

FAQs

- What types of documents can I upload?
 - The app supports PDF files and common image types like PNG and JPG.
- Do I need any programming knowledge?
 - No. The interface is designed for non-technical users to upload files, select a config, and view results.
- How do I know which config to choose?
 - Use the config that matches your document domain. For forms and receipts, choose the FUNSD-style config. For healthcare summary documents, choose the Healthcare SBC config.
- What does document validation mean here?
 - The system extracts key fields from each document and checks whether important values match between Document A and Document B.
- What if the results look incorrect?
 - The app may have trouble if the document scan is poor, the layout is unexpected, or the wrong config is selected. Try a clearer file or a more appropriate config.

Troubleshooting

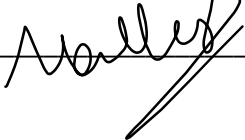
- If the upload fails:
 - Confirm the file is a supported PDF or image.
 - Try a smaller file or a different browser.
 - Refresh the page and try again.
- If extraction results are missing or incomplete:

- Check that you selected the correct config.
 - Use a cleaner scan or higher-quality image.
 - For PDFs, ensure the document text is legible.
- If validation shows unexpected mismatches:
 - Review whether Document A and Document B are meant to be compared.
 - Verify that both documents belong to the same domain and config.
 - Re-run the validation after retrying with the correct selection.
- If the app is slow or not responding:
 - Wait a few moments, as cloud model inference can take extra time.
 - Refresh the browser and try again.
 - Use a stable internet connection.
- If the live app is not reachable:
 - Wait a short time and retry later.
 - If you are running locally, make sure the backend and frontend are both started correctly.

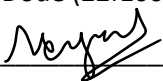
Team Member Confirmation

We confirm that we have reviewed the document and approve its submission.

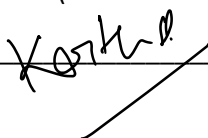
Name: Mallesh Mayara (21f2001118)

Signature: 

Name: Mayank Dode (22f1000781)

Signature: 

Name: Karthik Ganesh (21f2000775)

Signature: 

Name: Ayush Verma (21f3000500)

Signature: 